

ConstructionSceneReady: Evidence-grounded compilation and task-scene validation for BIM-driven robotic assembly

Zhu Aiyu^{a,*}

^a*Yango University, Quanzhou, 362014, China*

Abstract

Robot-ready construction simulation requires more than converting building geometry into a simulation format. We ask which building information actually supports a robotic construction task in simulation, and how its absence, error, or imprecision changes task outcomes. ConstructionSceneReady compiles IFC source into task-activated multi-scale OpenUSD scenes and validates them against an explicit task-scene contract with evidence-bounded repair. On 105 fault-injection cases the contract recovers all 141 findings without false positives and repairs 97.1% of scenes to pristine equivalence; an agent constrained to the same whitelist preserves this reliability, while an unconstrained agent produces validator-passing but incorrect scenes in 22.7%. Paired humanoid insertion episodes show symbolic and referential defects are discretely fatal, interface clearance is the most sensitive continuous property, and absolute member-pose errors are absorbed by closed-loop perception. Crossing these outcomes with the contract yields perfect specificity but limited geometric sensitivity, motivating two contract rules verified on an independent test set.

Acknowledgements

Not applicable.

*Corresponding author

Email address: `zhu.aiyu@yango.edu.cn` (Zhu Aiyu)

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.